

Nothing Lasts Forever – Only Item Administration: An Item Response Theory Algorithm to Shorten Tests

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Abstract. The responses to the items composing psychometrics tests are often used to estimate the respondents' latent trait levels. Although a larger number of items improves measurement validity, the effect of respondents' fatigue on the response quality should be acknowledged for developing reliable measurement tools. This contribution presents an item response theory-based algorithm (denoted as Léon) able to shorten existing tests by concurrently accounting for the measurement precision of the abbreviated test and response fatigue of the respondents. A simulation study compares the performance of Léon in approximating the measurement precision that would be obtained from the full-length test in absence of response fatigue against that of another algorithm that does not account for the response fatigue during the selection process. Although, on average, the two algorithms select the same number of items, Léon provides a better approximation to the measurement precision of the full-length test than the other algorithm. Limitations of the study are discussed.

Keywords: Item response theory, Careless error, Measurement precision, Short test forms

1 Introduction

As a general rule of thumb, the higher the number of items in a test, the better the measurement in terms of validity and reliability. However, there is a trade-off between the number of administered items and response quality. Therefore, it is important to balance the number of items with the potential for respondent fatigue in order to ensure reliable and precise measurements. Item Response Theory (IRT, see, e.g., [1]) provides an ideal framework for shortening existing tests given the detailed information on the measurement precision of each item with respect to different levels of the latent trait (see e.g., [3]). In this contribution, we present a new IRT-based algorithm for developing short test forms (STFs) from existing tests, denoted as Léon. This algorithm considers the response fatigue during the item selection process, such that it attempts at minimizing the number of selected items to maximize the measurement precision (as expressed by the test information function, TIF) of the STF.

2 Item Response Theory and Information Functions

In IRT models for dichotomous responses (e.g., correct vs. incorrect), the probability of observing a correct response on item i by person p depends on both the characteristics of the respondent (as described by their latent trait level, θ_p) and on the characteristics of the item. IRT models differentiate according to the number of parameters used for describing the characteristics of the items. According to the 4-Parameter logistic model (4-PL, [2]), the probability of a correct response is:

$$P(x_{pi} = 1 | \theta_p, b_i, a_i, c_i, d_i) = c_i + (d_i - c_i) \frac{\exp[a_i(\theta_p - b_i)]}{1 + \exp[a_i(\theta_p - b_i)]}, \quad (1)$$

where θ_p is the latent trait level of person p , b_i is the item location on the latent trait (i.e., difficulty parameter, the higher the value, the higher the difficulty of the item), and a_i is the item ability to discriminate between respondents with different latent trait levels (i.e., discrimination parameter, the higher the value, the higher the discrimination ability of the item). Parameters c_i and d_i are the lower and upper asymptotes of the probability, which should naturally tend to 0 and 1 when $\theta \rightarrow -\infty$ and $\theta \rightarrow +\infty$, respectively. The pseudoguessing parameter c_i is the probability of observing a correct response on i for $\theta \rightarrow -\infty$, such that the probability tends to c_i instead of 0. The d_i parameter is the probability of observing a correct response on i for $\theta \rightarrow +\infty$, such that this probability tends to d_i instead of 1. The careless error probability is $1 - d_i$ (i.e., the probability of not endorsing the item given that the $\theta_p > b_i$).

In this study, the response fatigue has been conceptualized as the careless error probability $1 - d_i$, which increases as the administration goes on. As $1 - d_i$ increases, the upper asymptote d_i decreases. This probability is intrinsically related to the item rank in the test, hence the careless error parameter in Equation (1), which is usually a property of the item (d_i), becomes a property of the item rank $r = \{1, \dots, R\}$ in the test, d_r , with the constraint $d_{r-1} > d_r$.

2.1 Information Functions

The measurement precision of each item with respect to different levels of the latent trait is expressed by the *item information functions* (IIFs). Given that in this application the response fatigue has been conceptualized as the careless errors as the administration goes on, the typical formalization of the IIF for the 4-PL [4] has been modified to account for the rank of presentation r of item i :

$$\text{IIF}_i(r) = \frac{a_i^2 [P(\theta) - c_i]^2 [d_r - P(\theta)]^2}{(d_r - c_i)^2 P(\theta) Q(\theta)}, \quad (2)$$

whereas in the typical formalization $d_r = d_i$. In Equation (2), $P(\theta) = P(x_{pi} = 1 | \theta, b_i, a_i, c_i, d_r)$ and $Q(\theta) = 1 - P(\theta)$. The IIF_i is strongly influenced by the item location on the latent trait with respect to a specific θ_p , its discriminativity, and its lucky guess and careless error. In absence of lucky guess and careless error

(i.e., $c_i = 0$, $d_i = 1$), the IIF is maximum when $\theta_p = b_i$, and it decreases as the distance between b_i and θ_p increases. When the lucky guess and careless error are taken into account, the informativeness of the item decreases, the IIF is lower, and its maximum is shifted.

The *test information function* (TIF) is the sum of the IIFs, such that its shape (i.e., its informativeness with respect to different latent trait levels) and its height (i.e., the amount of information for different latent trait levels) depend on the items distribution along the latent trait. The more the items with high discrimination and low lucky guess and careless error parameters are spread throughout the latent trait, the more the test is informative of different regions.

3 Item Selection Algorithms

The two algorithms (denoted as Frank and Léon) aim at reducing the distance between a target TIF (TIF-target, describes the desired measurement precision of the STF) and a provisional TIF (pTIF) obtained from the items included in the STF. At each iteration, an item is included in the STF according to its ability of bridging the gap between the two TIFs. The algorithms stop (i.e., termination criterion) when the distance between the TIF-target and the pTIF with the last considered item is equal to or greater than the distance between the TIF-target and the pTIF without the last considered items.

In what follows, the TIF-target is represented by the TIF that would be obtained if respondents never get tired during the administration of all the items in a test. **The set of items in the test is denoted as B .** Therefore, the TIF-target is denoted as TIF_B .

The set B includes the items without the response fatigue, $d_i = 1, \forall i \in B$. To include the response fatigue, the vector of careless error parameters d_r should be modified considering the item rank of presentation. In this application, d_r is modified with an exponential function, $d_r = \exp(-\lambda r)$ (where λ is the non-negative speed parameter determining the steepness of the function and r is the rank of the r -th item in the administration).

Since the TIF increases as the number of test items increases, the comparison between TIF_B and the TIF of the STFs is based on the mean TIF (i.e., the TIF divided by the number of items). In what follows, the mean TIF is referred to as TIF, $Q_x \subset B$, with $x \in \{\text{Frank}, \text{Léon}\}$ is the item subsets selected by Frank and Léon, and $||X||$ denotes the cardinality of set X .

Frank selects the item whose IIF is best able to reduce the distance from the TIF_B along the entire latent trait, without considering the respondents' fatigue, as follows:

At $k = 0$: $\text{TIF}^0(\theta) = 0 \forall \theta$, $Q^0 = \emptyset$. For $k \geq 0$,

1. $A^k = B \setminus Q^k$
2. $\forall i \in A^k$, $p\text{TIF}_i^k = \frac{\text{TIF}^k + \text{IIF}_i}{||Q^k|| + 1}$, with $c_i = 0$ and $d_i = 1, \forall i \in B$
3. $i^* = \arg \min_{i \in A^k} (|\text{TIF}_B - p\text{TIF}_i^k|)$
4. Termination criterion: $|\text{TIF}_B - p\text{TIF}_{i^*}^k| \geq |\text{TIF}_B - \text{TIF}^k|$:

- FALSE: $Q^{k+1} = Q^k \cup \{i^*\}$, $\text{TIF}^{k+1} = p\text{TIF}_{i^*}$, iterates 1-4
- TRUE: Stop, $Q_{\text{Frank}} = Q^k$

At $k = 0$, the subset of items Q^0 is empty and the TIF^0 is 0 for all the θ levels. At each iteration k : (1.) a set of available items is generated as the items in the item bank that have not been included in the STF yet, $A^k = B \setminus Q^k$; (2.) An average provisional TIF, $p\text{TIF}$, is computed by adding the IIF of each of the items in the set of the available items A^k , one at the time, to the TIF obtained from the items in Q^k (the denominator is obtained by adding 1 to the cardinality of Q^k); (3.) Among all the items in A^k , the one that allows for minimizing the distance between $p\text{TIF}$ and TIF_B is included in i^* ; (4.) The termination criterion is tested. If the distance between the TIF_B and the $p\text{TIF}_{i^*}$ is lower than the distance between the TIF_B and the TIF^k (i.e., the TIF obtained from the items in the subset Q^k , without item i^*) (FALSE), item i^* does contribute to reduce the distance from the TIF_B , hence it is included in the set of items and a new iteration starts, $Q^{k+1} = Q^k \cup \{i^*\}$. Conversely (TRUE), item i^* does not contribute in reducing the distance from the TIF_B , the algorithm stops, and the final item selection is the one without the item in i^* , $Q_{\text{Frank}} = Q^k$.

To account for the respondents' fatigue, at step (2.) Léon computes the IIF_i by considering the d_r related to the number of items included in the STF up to that point.

4 Simulation Study

4.1 Simulation Design

The performance of the algorithms is compared in 100 replications. At each replication, a test B composed of 50 items is generated by randomly drawing the difficulty and discrimination parameters from uniform distributions, $b_i \sim \mathcal{U}(-3, 3)$ and $a_i \sim \mathcal{U}(.90, 2)$, respectively. Lucky guess and careless error parameters are constant, $c_i = 0$ and $d_i = 1$, $\forall i \in B$. The TIF_B (i.e., the TIF-target) is obtained as the average TIF from the items in B , and describes the measurement precision that would be obtained if respondents were administered with all the items without getting tired. From the parameters of the items in B , a new test, B' , is generated, where $b'_i = b_i$, $a'_i = a_i$, $c_i = 0$, $\forall i \in B'$, and $d_r = \exp(-\lambda r)$, with $\lambda = 0.01$ and $r = \{0, \dots, ||B|| - 1\}$. An average TIF is computed considering the items in B' ($\text{TIF}_{B'}$), hence accounting for the response fatigue if all the items in the item bank are administered. Frank and Léon attempt at generating STFs able to approximate as best as possible TIF_B . Frank grounds the item selection on the set B , hence it does not consider the response fatigue during the selection process. To account for its effect, the TIFs of the STFs developed by Frank, $\text{TIF}_{\text{Frank}}$, are computed by considering the vector d_r associated to the rank of the items in Q_{Frank} . Grounding the item selection on B' , Léon already accounts for the effect of the response fatigue during the selection process itself. The TIFs obtained from the STFs developed by Léon are denoted as $\text{TIF}_{\text{Léon}}$.

4.2 Comparison

The average distance between TIF_B and $\text{TIF}_{\text{Frank}}$ and TIF_B and $\text{TIF}_{\text{Léon}}$ at each iteration, as well as the distance between TIF_B and $\text{TIF}_{B'}$, have been considered as a criterion for evaluating the ability of Léon of generating informative STFs while accounting for the response fatigue, $\Delta_x = |\text{TIF}_B - \text{TIF}_x|$, with $x \in \{B', \text{Frank}, \text{Léon}\}$. To better understand whether acknowledging the tiredness of the respondents influences the number of items included in Q , the cardinalities of the STFs generated by Frank and Léon, $||Q_{\text{Frank}}||$ and $||Q_{\text{Léon}}||$, respectively, have been compared.

5 Results

On average, Léon and Frank included the same number of items in the STF, $M_{\text{Léon}} = 6.17 \pm 3.34$, $M_{\text{Frank}} = 6.31 \pm 2.40$, suggesting that accounting for the tiredness of the respondents does not influence the number of items included in the STF. Figure 1 illustrates the distributions of the distances from TIF_B (y -axis) of B' , Frank, and Léon (x -axis).

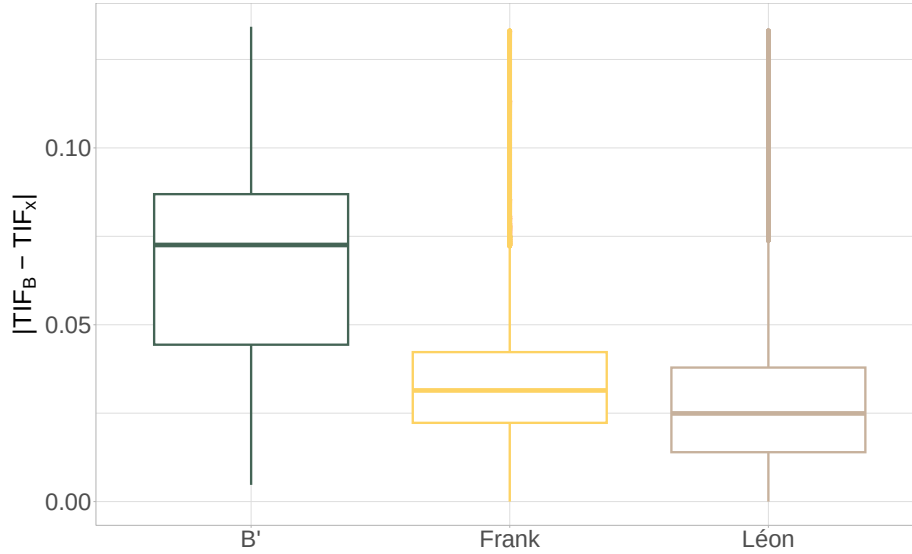


Fig. 1. Distance from TIF_B of $\text{TIF}_{B'}$, $\text{TIF}_{\text{Frank}}$, and $\text{TIF}_{\text{Léon}}$ across the 100 replications.

Overall, $\text{TIF}_{B'}$ s are the most distant from TIF_B , while $\text{TIF}_{\text{Léon}}$ s are the closest ones. Frank falls in between the two.

6 Final Remarks

This manuscript presented an IRT-based algorithm, denoted as Léon, for the generation of STFs. Léon grounds the item selection by concurrently considering the number of items included in the STF and the measurement precision with respect to an ideal target where all the items are administered in absence of response fatigue (i.e., TIF-target). The ability of Léon of approximating the TIF-target has been compared against that of another algorithm (Frank) not accounting for the response fatigue during item selection and with the measurement precision obtained from the administration of the entire test, including the careless mistakes due to the response fatigue.

Generally, the results suggest that the administration of fewer items provide better measurements than that of the entire test, especially if the selection process accounts for the effect of the response fatigue.

Although the results are promising, several limitations must be acknowledged. Firstly, while this study primarily focuses on approximating ideal measurement precision as expressed by the TIF-target, no investigations on the precision with which the latent trait is estimated with the STFs are presented. Future studies should further investigate this issue to understand whether administering less items brings actual benefits in the estimation of the latent trait. Secondly, the operationalization of the response fatigue as a decrease of the upper asymptote as the administration goes on is an arbitrary choice, and other possibilities should be investigated.

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